Lecture Notes: Graphs

Discrete Mathematics

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Contents

1. Basic Definitions

1.1. Graphs, Vertices, and Edges

Definition 1.1 (Graph)**.** A *graph* is an ordered pair $G = (V, E)$, where

- V is a non-empty set whose elements are called *vertices* (or *nodes*), and
- $E \subseteq \big\{\{u, v\} : u, v \in V,\ u \neq v\big\}$ is a set of *edges*, each of which is a two-element subset of V.

We write $|V|$ for the *order* of G and $|E|$ for its *size*.

Definition 1.2 (Incidence and Adjacency)**.** Let $G = (V, E)$ be a graph.

- Two vertices $u, v \in V$ are *adjacent* if $\{u, v\} \in E$.
- A vertex $v \in V$ and an edge $e \in E$ are *incident* if $v \in e$.
- Two edges $e_1, e_2 \in E$ are *adjacent* if they share a common vertex.

1.2. Types of Graphs

Definition 1.3 (Simple Graph)**.** A graph $G = (V, E)$ is called *simple* if it contains no *loops* (edges from a vertex to itself) and no *parallel edges* (multiple edges between the same pair of vertices). Definition 1.1 describes precisely the class of simple graphs.

Definition 1.4 (Multigraph)**.** A *multigraph* is a pair $G = (V, E)$ in which E is a *multiset* of two-element subsets of V, allowing two or more edges between the same pair of vertices (*parallel edges*). Loops may or may not be permitted depending on the convention adopted.

Definition 1.5 (Directed Graph)**.** A *directed graph* (or *digraph*) is an ordered pair $D = (V, A)$, where V is a vertex set and $A \subseteq V \times V$ is a set of *arcs* (directed edges). An arc (u, v) is distinct from (v, u).

Definition 1.6 (Weighted Graph)**.** A *weighted graph* is a triple $G = (V, E, w)$ where $G' = (V, E)$ is a graph (or digraph) and $w : E \to \mathbb{R}$ is a *weight function* assigning a real number to each edge.

1.3. Degree of a Vertex

Definition 1.7 (Degree)**.** Let $G = (V, E)$ be a simple graph. The *degree* of a vertex $v \in V$, denoted $\deg(v)$, is the number of edges incident to v:

$$\deg(v) = \big|\{e \in E : v \in e\}\big|. \tag{1}$$

For a digraph $D = (V, A)$:

- the *in-degree* of v is $\deg^-(v) = |\{(u, v) \in A\}|$,
- the *out-degree* of v is $\deg^+(v) = |\{(v, u) \in A\}|$.

Definition 1.8 (Degree Sequence)**.** The *degree sequence* of a graph G on n vertices is the non-decreasing sequence

$$(d_1, d_2, \ldots, d_n), \quad d_1 \leq d_2 \leq \cdots \leq d_n, \tag{2}$$

where $d_i = \deg(v_i)$ for each vertex $v_i \in V$.

1.4. Neighbourhood of a Vertex

Definition 1.9 (Neighbourhood)**.** Let $G = (V, E)$ be a graph and $v \in V$.

- The *(open) neighbourhood* of v is $\mathcal{N}(v) = \{u \in V : \{u, v\} \in E\}$.
- The *closed neighbourhood* of v is $\mathcal{N}[v] = \mathcal{N}(v) \cup \{v\}$.

Note that $\deg(v) = |\mathcal{N}(v)|$.

1.5. Illustrative Examples

Example 1.1 (A Simple Graph)**.** Consider the graph $G = (V, E)$ with $V = \{v_1, v_2, v_3, v_4, v_5\}$ and $E = \{\{v_1, v_2\}, \{v_1, v_3\}, \{v_2, v_4\}, \{v_3, v_4\}, \{v_4, v_5\}\}$. This graph is depicted in Figure 1. Its degree sequence is $(1, 2, 2, 2, 3)$.

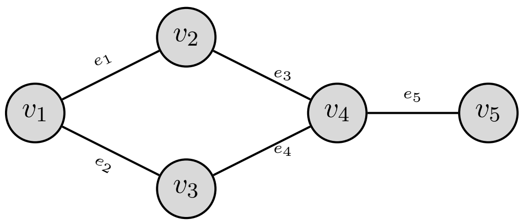

Figure 1: A simple graph G on five vertices and five edges, as described in Example 1.1. Edge labels $e_1, \ldots, e_5$ are shown for reference.

Example 1.2 (A Directed Graph)**.** Consider the digraph $D = (V, A)$ with $V = \{u_1, u_2, u_3, u_4\}$ and $A = \{(u_1, u_2), (u_2, u_3), (u_3, u_1), (u_1, u_4), (u_4, u_3)\}$. This digraph is depicted in Figure 2. We observe $\deg^-(u_1) = 1$, $\deg^+(u_1) = 2$.

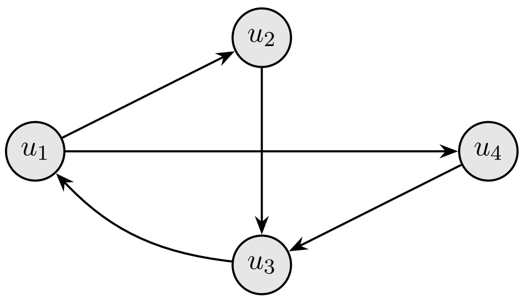

Figure 2: A directed graph (digraph) D on four vertices and five arcs, as described in Example 1.2.

2. Basic Properties of Graphs

2.1. The Handshaking Lemma

Theorem 2.1 (Handshaking Lemma)**.** *Let $G = (V, E)$ be a simple graph. Then*

$$\sum_{v \in V} \deg(v) = 2|E|. \tag{3}$$

Proof. Each edge $e = \{u, v\} \in E$ is incident to exactly two vertices, namely u and v. Therefore, when summing the degrees over all vertices, every edge is counted exactly twice — once for each of its two endpoints. Formally, by interchanging the order of summation:

$$\sum_{v\in V} \deg(v) = \sum_{v\in V} \big|\{e \in E : v \in e\}\big| = \sum_{e\in E} \big|\{v \in V : v \in e\}\big| = \sum_{e\in E} 2 = 2|E|. \tag{4}$$

□

Corollary 2.1.1. *In any graph $G = (V, E)$, the number of vertices of odd degree is even.*

Proof. Partition V into $V_{\text{odd}} = \{v \in V : \deg(v) \text{ is odd}\}$ and $V_{\text{even}} = V \setminus V_{\text{odd}}$. By Theorem 2.1,

$$\sum_{v\in V_{\text{odd}}} \deg(v) + \sum_{v\in V_{\text{even}}} \deg(v) = 2|E|. \tag{5}$$

The right-hand side and the second summand on the left are both even, hence $\sum_{v\in V_{\text{odd}}} \deg(v)$ is even. Since each term in this sum is odd, the number of terms $|V_{\text{odd}}|$ must be even. □

2.2. Special Vertices

Definition 2.1 (Isolated, Pendant, and Dominating Vertices)**.** Let $G = (V, E)$ be a graph.

- A vertex $v \in V$ is *isolated* if $\deg(v) = 0$, i.e. $\mathcal{N}(v) = \emptyset$.
- A vertex $v \in V$ is a *pendant vertex* (or *leaf*) if $\deg(v) = 1$.
- A vertex $v \in V$ is *dominating* if $\mathcal{N}[v] = V$, i.e. v is adjacent to every other vertex.

2.3. Walks, Trails, Paths, and Cycles

Definition 2.2 (Walk)**.** A *walk* of *length* k in a graph $G = (V, E)$ is a finite sequence

$$W = v_0,\ e_1,\ v_1,\ e_2,\ v_2,\ \dots,\ e_k,\ v_k, \tag{6}$$

where $v_i \in V$ for each i and $e_j = \{v_{j-1}, v_j\} \in E$ for each j. The walk is *closed* if $v_0 = v_k$.

Definition 2.3 (Trail and Path)**.** Let W be a walk as in Definition 2.2.

- W is a *trail* if all edges $e_1, \dots, e_k$ are distinct.
- W is a *path* if additionally all vertices $v_0, v_1, \dots, v_k$ are distinct. A path from u to v is denoted u-v path.

Definition 2.4 (Cycle)**.** A *cycle* of length $k \geq 3$ is a closed walk $v_0, e_1, v_1, \dots, e_k, v_0$ in which all edges are distinct and all vertices $v_0, v_1, \dots, v_{k-1}$ are distinct. A cycle of length k is denoted C_k.

2.4. Connectedness and Components

Definition 2.5 (Connected Graph)**.** A graph $G = (V, E)$ is *connected* if for every pair of distinct vertices $u, v \in V$ there exists a u-v path in G. Otherwise G is *disconnected*.

Definition 2.6 (Connected Component)**.** A *connected component* of a graph G is a maximal connected subgraph of G. Every graph is the disjoint union of its connected components.

2.5. Subgraphs

Definition 2.7 (Subgraph)**.** A graph $H = (V', E')$ is a *subgraph* of $G = (V, E)$, written $H \subseteq G$, if $V' \subseteq V$ and $E' \subseteq E$.

- H is a *spanning subgraph* of G if $V' = V$ (the vertex set is preserved).
- H is the *subgraph induced* by $S \subseteq V$, written $G[S]$, if $V' = S$ and $E' = \{\{u, v\} \in E : u, v \in S\}$ (all edges of G with both endpoints in S are retained).

2.6. Illustrative Example: Paths and Cycles

Example 2.1 (Path and Cycle)**.** Figure 3 illustrates a graph G on six vertices containing both a highlighted path $P = v_1, v_2, v_3, v_4$ (shown in blue) and a highlighted cycle $C = v_3, v_4, v_5, v_6, v_3$ (shown in red).

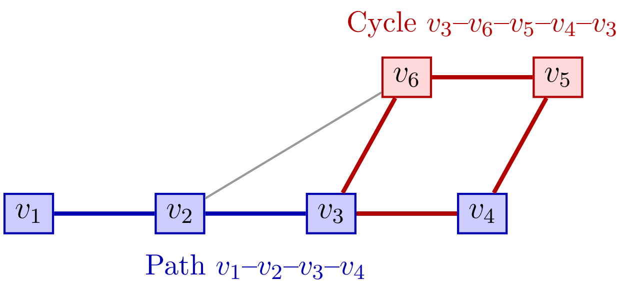

Figure 3: A graph illustrating a path (blue) and a cycle (red) sharing the edge $\{v_3, v_4\}$, as described in Example 2.1.

Remark 2.1. Every cycle C_k ($k \geq 3$) is a connected graph in which every vertex has degree exactly 2. Conversely, a connected graph in which every vertex has degree 2 is a cycle.

3. Graph Isomorphism

3.1. Motivation

Two graphs may appear different — possessing distinct vertex labels or geometric representations — yet share an identical abstract structure. The notion of *graph isomorphism* makes this intuition precise by requiring the existence of a structure-preserving bijection between vertex sets.

Remark 3.1. When studying graph-theoretic properties (connectivity, degree sequences, cycle lengths, etc.), the actual names of vertices are irrelevant. Two isomorphic graphs satisfy exactly the same graph-theoretic statements.

3.2. Formal Definition

Definition 3.1 (Graph Isomorphism)**.** Two graphs $G_1 = (V_1, E_1)$ and $G_2 = (V_2, E_2)$ are *isomorphic*, written $G_1 \cong G_2$, if there exists a bijection

$$f : V_1 \longrightarrow V_2 \tag{7}$$

such that for all $u, v \in V_1$,

$$\{u, v\} \in E_1 \iff \{f(u), f(v)\} \in E_2. \tag{8}$$

Such a bijection f is called an *isomorphism* from G_1 to G_2.

Remark 3.2. Graph isomorphism is an equivalence relation on the class of all graphs: it is reflexive (the identity map is always an isomorphism $G \cong G$), symmetric (if f is an isomorphism $G_1 \to G_2$ then f^{-1} is an isomorphism $G_2 \to G_1$), and transitive (the composition of two isomorphisms is an isomorphism).

3.3. Graph Invariants

A *graph invariant* is a property that takes the same value on isomorphic graphs. Invariants provide a practical tool for proving non-isomorphism: if two graphs differ on any invariant, they cannot be isomorphic.

Definition 3.2 (Graph Invariants)**.** The following are well-known graph invariants:

1. **Order and size:** $|V|$ and $|E|$.
2. **Degree sequence:** the non-decreasing sequence of vertex degrees (Definition 1.8).
3. **Cycle spectrum:** the set of lengths of cycles present in the graph.
4. **Connectivity:** whether the graph is connected; the number of connected components.
5. **Diameter:** $\operatorname{diam}(G) = \max_{u,v \in V} d(u, v)$, where $d(u, v)$ denotes the length of a shortest u-v path.

Theorem 3.1 (Necessary Conditions for Isomorphism)**.** *If $G_1 \cong G_2$ via isomorphism f, then:*

1. $|V_1| = |V_2|$ *and* $|E_1| = |E_2|$,
2. G_1 *and* G_2 *have the same degree sequence,*
3. G_1 *is connected if and only if* G_2 *is connected.*

Proof. Since $f : V_1 \to V_2$ is a bijection, $|V_1| = |V_2|$. Because f preserves adjacency (equation (8)), it induces a bijection on edge sets, whence $|E_1| = |E_2|$. For each vertex $v \in V_1$, adjacency-preservation implies $\deg_{G_1}(v) = \deg_{G_2}(f(v))$, so the degree sequences coincide. Connectivity is preserved because f maps paths in G_1 to paths in G_2. $\square$

Remark 3.3. The conditions in Theorem 3.1 are *necessary* but not *sufficient.* Two graphs may share the same order, size, and degree sequence while failing to be isomorphic. Determining whether two graphs are isomorphic is a computationally challenging problem (graph isomorphism lies in NP and is not known to be NP-complete).

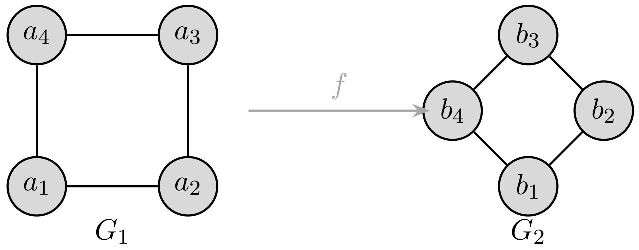

Figure 4: Two isomorphic graphs $G_1 \cong G_2$ (both are C_4) with different vertex labels and geometric embeddings, as described in Example 3.1.

3.4. Examples

Example 3.1 (Two Isomorphic Graphs)**.** The graphs G_1 and G_2 depicted in Figure 4 are isomorphic. Both have order 4, size 4, and degree sequence $(2, 2, 2, 2)$. The isomorphism

$$f(a_1) = b_1, \quad f(a_2) = b_2, \quad f(a_3) = b_3, \quad f(a_4) = b_4 \tag{9}$$

witnesses $G_1 \cong G_2$, as may be verified by checking that each edge of G_1 maps to an edge of G_2 and vice versa.

Example 3.2 (Two Non-Isomorphic Graphs)**.** The graphs H_1 and H_2 depicted in Figure 5 both have order 4 and size 4. However, their degree sequences differ:

$$\text{degree sequence of } H_1 = (1, 1, 2, 4), \tag{10}$$
$$\text{degree sequence of } H_2 = (2, 2, 2, 2). \tag{11}$$

By Theorem 3.1, $H_1 \ncong H_2$.

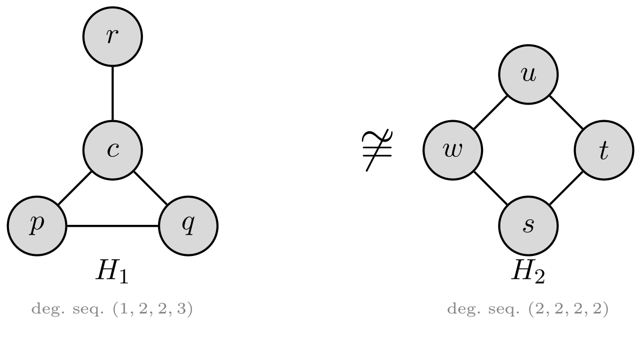

Figure 5: Two non-isomorphic graphs $H_1 \ncong H_2$ on four vertices and four edges, distinguished by their degree sequences (see Example 3.2).

Remark 3.4. Although the degree sequence is a convenient invariant for refuting isomorphism (as in Example 3.2), no finite set of polynomial-time invariants is known that *decides* isomorphism for all graphs. The graph isomorphism problem remains one of the most intriguing open problems at the interface of combinatorics and computational complexity.